

d) $E_i = \text{grade on problem } i$

$$E = \sum_{i=1}^{20} E_i$$

$$E[E] = E\left[\sum_{i=1}^{20} E_i\right] = \sum_{i=1}^{20} E[E_i] = 20 \cdot 0.0468 \\ = 0.936$$

e) Part a has higher variances, this is because part b had more values comparing that when averaged out, makes the variance smaller. More extreme scores are "weighed" less when larger amounts of data is present.